

National Neighborhood Data Archive (NaNDA): Post Offices and Banks by Census Tract, United States, 2003-2017



openICPSR-128281

nanda_pobank_tract_2003-2017_02P.dta

nanda_pobank_tract_2003-2017_02P.csv

nanda_pobank0317T_02P.sas7bdat

Overview and Data Dictionary

Documentation Version: 2.0

Last updated: 10/21/2022

Dataset Overview

Description

This dataset contains measures of the number and per capita density of post offices and banks per United States census tract from 2003 through 2017.

Principal Investigators

- Jessica Finlay, University of Michigan Institute for Social Research
- Mao Li, University of Michigan Institute for Social Research
- Michael Esposito, University of Michigan Institute for Social Research
- Iris Gomez-Lopez, University of Michigan Institute for Social Research
- Anam Khan, University of Michigan Institute for Social Research
- Philippa Clarke, University of Michigan Institute for Social Research
- Megan Chenoweth, University of Michigan Institute for Social Research

Funding Sources

- United States Department of Health and Human Services. Administration for Community Living. National Institute on Disability, Independent Living, and Rehabilitation Research (90RTHF0001)
- United States Department of Health and Human Services. National Institutes of Health. National Institute on Aging (RF1-AG-057540)

Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations of interest. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, including Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the concentration of post offices and banks per census tract in the United States. These businesses, like other neighborhood retail establishments, function as common “third places” in the United States alongside more common gathering places (e.g. coffee shops, churches, gyms, senior centers). Third places are locations other than home (first place) and

work/school (second place) where people gather, spend time, and casually interact with one another in ways that foster social interaction and community building. They are social infrastructure vital to civic life (Klinenberg, 2018), and their presence or absence in a neighborhood has affects individual and community well-being (Finlay et al., 2018).

Our research team identified the following NAICS codes of interest:

- 491: Post offices and other United States Postal Service facilities.
- 522110: Commercial banks
- 522120: Savings and loan banks.
- 522130: Credit unions.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment's census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all post offices and banks per tract in each year.

Usage Note

Data users interested in the broader retail environment might find further useful data in these additional NaNDA datasets:

- [Dollar stores by census tract, United States, 2003-2017](#) (Gomez-Lopez et al., 2020).
- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020a), which includes coffee shops, fast food restaurants, and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020b).
- [Health care services by census tract, United States, 2003-2017](#) (Khan et al., 2020), which includes pharmacies and stores selling optical goods and health supplies.
- [Liquor, tobacco, and convenience stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020c).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Esposito et al., 2020b), which includes barber shops and hair and nail salons.
- [Retail establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020e), which includes department, furniture, home and garden, clothing, and thrift stores, among others.

Those interested in third places and community resources might find further useful data in these additional NaNDA datasets which are not focused on the retail environment:

- [Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020a), which includes gyms, libraries, and arcades, among others.
- [Religious, civic, and social organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020d).
- [Social services by census tract, United States, 2003-2017](#) (Finlay et al., 2020f), which includes senior centers.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Variable label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_491	int	1036462	22	0.4274966	0	56	# post offices/postal service facilities
count_sales_491	int	1036462	9	0.023805	0	9	# post offices/postal service facilities w/ sales>0
count_emp_491	int	1036462	22	0.389145	0	54	# post offices/postal service facilities w/ 2+ employees
popden_491	float	1015404	120115	30.08684	-1	1.17E+07	# post offices/postal service facilities per 1000 people
popden_sales_491	float	1015105	12932	0.0237169	-1	3333.333	# post offices/postal service facilities per 1000 people w/ sales>0
popden_emp_491	float	1015385	114648	30.05593	-1	1.17E+07	# post offices/postal service facilities per 1000 people w/ 2+ employees
aden_491	float	1017996	30708	0.1889786	0	116.9832	# post offices/postal service facilities per sq mile
aden_sales_491	float	1017996	4215	0.01571	0	42.71793	# post offices/postal service facilities per sq mile w/ sales>0
aden_emp_491	float	1017996	30618	0.1779166	0	112.8053	# post offices/postal service facilities per sq mile w/ 2+ employees
count_522110	int	1036462	91	1.11854	0	225	# commercial banks
count_sales_522110	int	1036462	89	1.060566	0	220	# commercial banks w/ sales>0
count_emp_522110	int	1036462	93	1.097893	0	222	# commercial banks w/ 2+ employees
popden_522110	float	1015472	181090	0.5732128	-1	19946.46	# commercial banks per 1000 people
popden_sales_522110	float	1015453	177537	0.5506012	-1	19946.46	# commercial banks per 1000 people w/ sales>0
popden_emp_522110	float	1015464	179322	0.562229	-1	19946.46	# commercial banks per 1000 people w/ 2+ employees

National Neighborhood Data Archive (NaNDA)
Post Offices and Banks by Census Tract, United States, 2003-2017

Variable	Type	Obs	Unique	Mean	Min	Max	Variable label
aden_522110	float	1017996	85192	1.189644	0	1051.228	# commercial banks per sq mile
aden_sales_522110	float	1017996	83865	1.120134	0	1024.734	# commercial banks per sq mile w/ sales>0
aden_emp_522110	float	1017996	84758	1.168394	0	1034.05	# commercial banks per sq mile w/ 2+ employees
count_522120	int	1036462	21	0.2104853	0	30	# savings and loan banks
count_sales_522120	int	1036462	22	0.201087	0	29	# savings and loan banks w/ sales>0
count_emp_522120	int	1036462	22	0.2062034	0	30	# savings and loan banks w/ 2+ employees
popden_522120	float	1015148	71079	0.1210564	-1	12501.77	# savings and loan banks per 1000 people
popden_sales_522120	float	1015144	69209	0.11329	-1	12501.77	# savings and loan banks per 1000 people w/ sales>0
popden_emp_522120	float	1015148	70141	0.1191524	-1	12501.77	# savings and loan banks per 1000 people w/ 2+ employees
aden_522120	float	1017996	22843	0.2736186	0	160.495	# savings and loan banks per sq mile
aden_sales_522120	float	1017996	22546	0.2607098	0	144.4455	# savings and loan banks per sq mile w/ sales>0
aden_emp_522120	float	1017996	22599	0.2680625	0	144.4455	# savings and loan banks per sq mile w/ 2+ employees
count_522130	int	1036462	21	0.2396364	0	21	# credit unions
count_sales_522130	int	1036462	19	0.2276292	0	18	# credit unions w/ sales>0
count_emp_522130	int	1036462	20	0.2278945	0	21	# credit unions w/ 2+ employees
popden_522130	float	1015412	76001	1.49595	-1	507204.4	# credit unions per 1000 people
popden_sales_522130	float	1015404	73882	1.487662	-1	507204.4	# credit unions per 1000 people w/ sales>0
popden_emp_522130	float	1015383	73390	1.483654	-1	507204.4	# credit unions per 1000 people w/ 2+ employees
aden_522130	float	1017996	24595	0.243197	0	80.57078	# credit unions per sq mile
aden_sales_522130	float	1017996	24206	0.2308837	0	80.57078	# credit unions per sq mile w/ sales>0
aden_emp_522130	float	1017996	24012	0.2300957	0	80.57078	# credit unions per sq mile w/ 2+ employees

References

- Clarke, P. J., & Melendez, R. (2019). National Neighborhood Data Archive (NaNDA): Neighborhood socioeconomic and demographic characteristics of census tracts, United States, 2000-2010 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E111107V1>
- Esposito, M., Li, M., Finlay, J. M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., and Chenoweth, M. M. (2020a). National Neighborhood Data Archive (NaNDA): Eating and drinking places by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E115404V2>
- Esposito, M., Li, M., Finlay, J. M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., and Chenoweth, M. M. (2020b). National Neighborhood Data Archive (NaNDA): Personal care services and laundromats by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E115981V2>
- Finlay, J. M., Gaugler, J. E., & Kane, R. L. (2018). Ageing in the margins: Expectations of and struggles for “a good place to grow old” among low-income older Minnesotans. *Ageing and Society*, 1-25. <https://doi.org/10.1017/S0144686X1800123X>
- Finlay, J. M., Esposito, M., Kim, M. H., Gomez-Lopez, I. N., & Clarke, P. J. (2019). Closure of “third places”? Exploring potential consequences for collective health and wellbeing. *Health & Place*, 60, 102225. <https://doi.org/10.1016/j.healthplace.2019.102225>
- Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020a). National Neighborhood Data Archive (NaNDA): Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E115543V2>
- Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020b). National Neighborhood Data Archive (NaNDA): Grocery stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E123001V1>
- Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020c). National Neighborhood Data Archive (NaNDA): Liquor, tobacco, and convenience stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E123541V1>
- Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020d). National Neighborhood Data Archive (NaNDA): Religious, civic, and social organizations by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E115967V1>

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020e). National Neighborhood Data Archive (NaNDA): Retail establishments by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research.

<https://doi.org/10.3886/E115972V2>

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020f). National Neighborhood Data Archive (NaNDA): Social services by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research.

<https://doi.org/10.3886/E117163V2>

Gomez-Lopez, I. N., Clarke, P. J., Hill, A. B., Romero, D. M., Goodspeed, R., Berrocal, V. J., Vinod Vydiswaran, V.G, & Veinot, T. C. (2017). Using social media to identify sources of healthy food in urban neighborhoods. *Journal of Urban Health*, 94(3), 429–36. <https://doi.org/10.1007/s11524-017-0154-1>

Gomez-Lopez, I. N., Esposito, M., Li, M., Finlay, J. M., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Dollar stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research.

<https://doi.org/10.3886/E123802V1>

Khan, A. M., Li, M., Finlay, J. M., Esposito, M., Gomez-Lopez, I. N., Clarke, P. J., & Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Health care services by census tract, United States, 2008-2017 (V2) [Data set]. <https://doi.org/10.3886/E120907V2>

Klinenberg, E. (2018). *Palaces for the people: How social infrastructure can help fight inequality, polarization, and the decline of civic life*. Broadway Books.

Melendez, R., Clarke, P. J., Khan, A. M., Gomez-Lopez, I. N., Li, M., & Chenoweth, M. (2020). National Neighborhood Data Archive (NaNDA): Socioeconomic status and demographic characteristics of census tracts, United States, 2008-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E120462V1>

United States Census Bureau (2010). TIGER/Line shapefiles, 2010 census tracts (2010 version) [Data set]. https://www2.census.gov/geo/tiger/TIGER2010/TRACT/2010/tl_2010_us_tract10.zip.

United States Census Bureau (2017). North American Industry Classification System (2017 version) [Data set]. <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>

Walls & Associates (2017). National Establishment Time Series (NETS) database.